

APM1220: FA 2 Solutions

Problem 3.1

Given:

$$X = \begin{bmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{bmatrix}$$

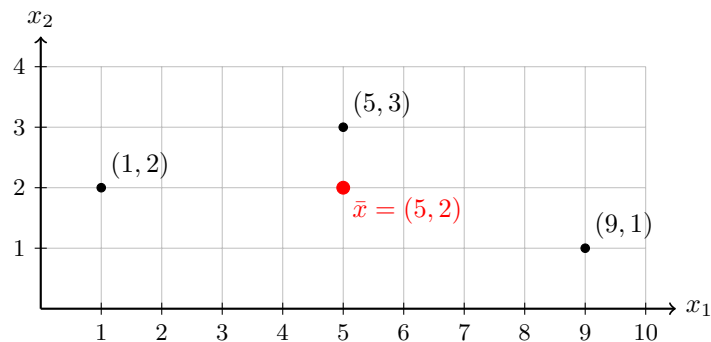
(a) Scatter plot and sample mean

First, calculate the sample mean $\bar{x} = [\bar{x}_1, \bar{x}_2]'$:

$$\bar{x}_1 = \frac{9 + 5 + 1}{3} = \frac{15}{3} = 5$$

$$\bar{x}_2 = \frac{1 + 3 + 2}{3} = \frac{6}{3} = 2$$

The sample mean is $\bar{x} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$.



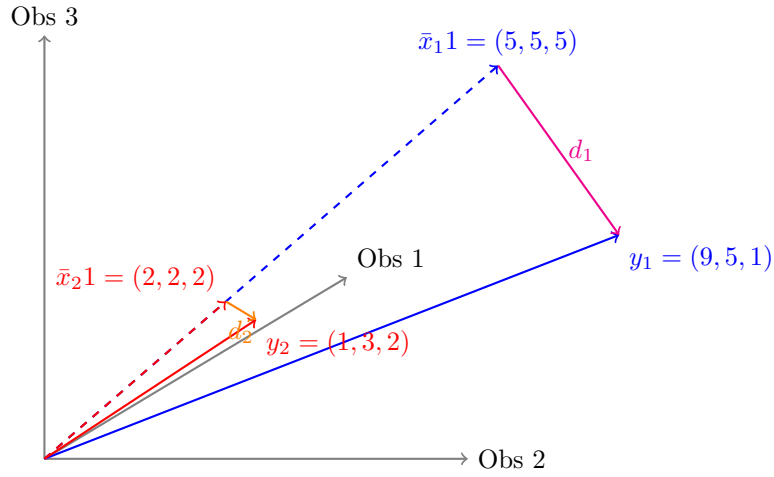
(b) Sketch the $n = 3$ dimensional representation and deviation vectors

Calculate the deviation vectors $d_1 = y_1 - \bar{x}_1 1$ and $d_2 = y_2 - \bar{x}_2 1$:

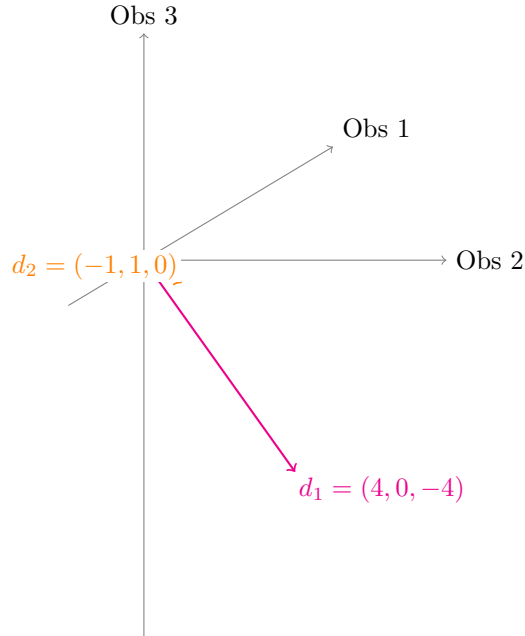
$$y_1 = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix}, \quad \bar{x}_1 1 = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} \implies d_1 = \begin{bmatrix} 9 - 5 \\ 5 - 5 \\ 1 - 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \quad \bar{x}_2 1 = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \implies d_2 = \begin{bmatrix} 1 - 2 \\ 3 - 2 \\ 2 - 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Here is the 3D sketch showing the original data points (y_1, y_2) , their corresponding mean vectors $(\bar{x}_1 1, \bar{x}_2 1)$, and the deviation gap between them:



(c) Deviation vectors from the origin, lengths, angles, S_n , and R
 Here are the deviation vectors d_1 and d_2 emanating directly from the origin:



Calculate the lengths of the deviation vectors (L_{d_1} , L_{d_2}):

$$L_{d_1} = \sqrt{4^2 + 0^2 + (-4)^2} = \sqrt{16 + 0 + 16} = \sqrt{32} = 4\sqrt{2}$$

$$L_{d_2} = \sqrt{(-1)^2 + 1^2 + 0^2} = \sqrt{1 + 1 + 0} = \sqrt{2}$$

Calculate the dot product $d_1 \cdot d_2$:

$$d_1 \cdot d_2 = (4)(-1) + (0)(1) + (-4)(0) = -4 + 0 + 0 = -4$$

Calculate the cosine of the angle θ :

$$\cos(\theta) = \frac{d_1 \cdot d_2}{L_{d_1} L_{d_2}} = \frac{-4}{(\sqrt{32})(\sqrt{2})} = \frac{-4}{\sqrt{64}} = \frac{-4}{8} = -0.5$$

Relate to S_n (sample covariance matrix with divisor $n = 3$):

$$S_n = \frac{1}{n} \begin{bmatrix} L_{d_1}^2 & d_1 \cdot d_2 \\ d_1 \cdot d_2 & L_{d_2}^2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 32 & -4 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} 32/3 & -4/3 \\ -4/3 & 2/3 \end{bmatrix}$$

Relate to R (sample correlation matrix). The correlation coefficient r_{12} is exactly $\cos(\theta)$:

$$R = \begin{bmatrix} 1 & \cos(\theta) \\ \cos(\theta) & 1 \end{bmatrix} = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$

Problem 3.6

Given:

$$X = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix}$$

(a) Matrix of deviations $(X - 1\bar{x}')$

First, find the sample means for each column:

$$\bar{x}_1 = \frac{-1 + 2 + 5}{3} = \frac{6}{3} = 2$$

$$\bar{x}_2 = \frac{3 + 4 + 2}{3} = \frac{9}{3} = 3$$

$$\bar{x}_3 = \frac{-2 + 2 + 3}{3} = \frac{3}{3} = 1$$

Construct the mean matrix $1\bar{x}'$:

$$1\bar{x}' = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 1 \\ 2 & 3 & 1 \\ 2 & 3 & 1 \end{bmatrix}$$

Calculate the deviation matrix $D = X - 1\bar{x}'$:

$$D = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 3 & 1 \\ 2 & 3 & 1 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix}$$

Is it full rank? No. The sum of the columns in a deviation matrix is always zero. Specifically, row 1 plus row 2 plus row 3 equals exactly the zero vector $[0, 0, 0]$. Since the rows are linearly dependent, the rank is at most 2, meaning it is not full rank (rank < 3).

(b) Sample Covariance S and Generalized Variance $|S|$

$$\begin{aligned} D'D &= \begin{bmatrix} -3 & 0 & 3 \\ 0 & 1 & -1 \\ -3 & 1 & 2 \end{bmatrix} \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} (-3)(-3) + (0)(0) + (3)(3) & (-3)(0) + (0)(1) + (3)(-1) & (-3)(-3) + (0)(1) + (3)(2) \\ (0)(-3) + (1)(0) + (-1)(3) & (0)(0) + (1)(1) + (-1)(-1) & (0)(-3) + (1)(1) + (-1)(2) \\ (-3)(-3) + (1)(0) + (2)(3) & (-3)(0) + (1)(1) + (2)(-1) & (-3)(-3) + (1)(1) + (2)(2) \end{bmatrix} \\ &= \begin{bmatrix} 9 + 0 + 9 & 0 + 0 - 3 & 9 + 0 + 6 \\ 0 + 0 - 3 & 0 + 1 + 1 & 0 + 1 - 2 \\ 9 + 0 + 6 & 0 + 1 - 2 & 9 + 1 + 4 \end{bmatrix} = \begin{bmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{bmatrix} \end{aligned}$$

Calculate $S = \frac{1}{n-1} D'D = \frac{1}{2} D'D$:

$$S = \frac{1}{2} \begin{bmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{bmatrix} = \begin{bmatrix} 9 & -3/2 & 15/2 \\ -3/2 & 1 & -1/2 \\ 15/2 & -1/2 & 7 \end{bmatrix}$$

Generalized Variance $|S|$: Because the deviation matrix is not full rank, the covariance matrix S is singular. Therefore, $|S| = 0$.

Geometric Interpretation: The generalized variance represents the 3D volume formed by the three

deviation vectors. Because the vectors are linearly dependent, they lie completely flat on a 2D plane. Since a flat shape has no 3D volume, the generalized variance is zero.

(c) Total Sample Variance

Total variance is the trace of S (sum of diagonal elements):

$$Tr(S) = s_{11} + s_{22} + s_{33} = 9 + 1 + 7 = 17$$

Problem 3.9

Given:

$$X = \begin{bmatrix} 12 & 17 & 29 \\ 18 & 20 & 38 \\ 14 & 16 & 30 \\ 20 & 18 & 38 \\ 16 & 19 & 35 \end{bmatrix}$$

(a) Mean corrected matrix D and linear dependence

Calculate the column means:

$$\bar{x}_1 = \frac{12 + 18 + 14 + 20 + 16}{5} = \frac{80}{5} = 16$$

$$\bar{x}_2 = \frac{17 + 20 + 16 + 18 + 19}{5} = \frac{90}{5} = 18$$

$$\bar{x}_3 = \frac{29 + 38 + 30 + 38 + 35}{5} = \frac{170}{5} = 34$$

Subtract means to get $D = X - 1\bar{x}'$:

$$D = \begin{bmatrix} 12 - 16 & 17 - 18 & 29 - 34 \\ 18 - 16 & 20 - 18 & 38 - 34 \\ 14 - 16 & 16 - 18 & 30 - 34 \\ 20 - 16 & 18 - 18 & 38 - 34 \\ 16 - 16 & 19 - 18 & 35 - 34 \end{bmatrix} = \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix}$$

The vector establishing linear dependence is $a = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$. Verifying $Da = 0$:

$$Da = \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -4(1) - 1(1) - 5(-1) \\ 2(1) + 2(1) + 4(-1) \\ -2(1) - 2(1) - 4(-1) \\ 4(1) + 0(1) + 4(-1) \\ 0(1) + 1(1) + 1(-1) \end{bmatrix} = \begin{bmatrix} -5 + 5 \\ 4 - 4 \\ -4 + 4 \\ 4 - 4 \\ 1 - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

(b) Sample Covariance S and generalized variance

Calculate $D'D$:

$$D'D = \begin{bmatrix} -4 & 2 & -2 & 4 & 0 \\ -1 & 2 & -2 & 0 & 1 \\ -5 & 4 & -4 & 4 & 1 \end{bmatrix} \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 40 & 12 & 52 \\ 12 & 10 & 22 \\ 52 & 22 & 74 \end{bmatrix}$$

Calculate $S = \frac{1}{n-1} D'D = \frac{1}{4} D'D$:

$$S = \frac{1}{4} \begin{bmatrix} 40 & 12 & 52 \\ 12 & 10 & 22 \\ 52 & 22 & 74 \end{bmatrix} = \begin{bmatrix} 40/4 & 12/4 & 52/4 \\ 12/4 & 10/4 & 22/4 \\ 52/4 & 22/4 & 74/4 \end{bmatrix} = \begin{bmatrix} 10 & 3 & 13 \\ 3 & 5/2 & 11/2 \\ 13 & 11/2 & 37/2 \end{bmatrix}$$

Because the columns are linearly dependent, the matrix is singular. Thus, $|S| = 0$. Verify $Sa = 0$:

$$Sa = \begin{bmatrix} 10 & 3 & 13 \\ 3 & 5/2 & 11/2 \\ 13 & 11/2 & 37/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 10(1) + 3(1) + 13(-1) \\ 3(1) + 5/2(1) + 11/2(-1) \\ 13(1) + 11/2(1) + 37/2(-1) \end{bmatrix} = \begin{bmatrix} 13 - 13 \\ 11/2 - 11/2 \\ 37/2 - 37/2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(c) Third column sum verification

The relationship $a_1x_1 + a_2x_2 + a_3x_3 = 0$ with $a_1 = 1, a_2 = 1, a_3 = -1$ yields:

$$(1)x_1 + (1)x_2 + (-1)x_3 = 0 \implies x_1 + x_2 - x_3 = 0 \implies x_3 = x_1 + x_2$$

This algebraic equivalence proves the third column is strictly the sum of the first two columns.

Problem 3.18

Given:

$$\bar{x} = \begin{bmatrix} 0.766 \\ 0.508 \\ 0.438 \\ 0.161 \end{bmatrix}, \quad S = \begin{bmatrix} 0.856 & 0.635 & 0.173 & 0.096 \\ 0.635 & 0.568 & 0.128 & 0.067 \\ 0.173 & 0.127 & 0.171 & 0.039 \\ 0.096 & 0.067 & 0.039 & 0.043 \end{bmatrix}$$

(a) Sample mean and variance of total energy consumption

Total consumption is $c'x$ where $c = [1 \quad 1 \quad 1 \quad 1]'$.

$$\text{Mean} = c'\bar{x} = 0.766 + 0.508 + 0.438 + 0.161 = 1.873$$

The variance is $c'Sc$, which equals the sum of all elements in S . Calculate row sums first:

$$\text{Row 1 Sum} = 0.856 + 0.635 + 0.173 + 0.096 = 1.760$$

$$\text{Row 2 Sum} = 0.635 + 0.568 + 0.128 + 0.067 = 1.398$$

$$\text{Row 3 Sum} = 0.173 + 0.127 + 0.171 + 0.039 = 0.510$$

$$\text{Row 4 Sum} = 0.096 + 0.067 + 0.039 + 0.043 = 0.245$$

$$\text{Variance} = 1.760 + 1.398 + 0.510 + 0.245 = 3.913$$

(b) Excess of petroleum over natural gas

This difference is $b'x$ where $b = [1 \quad -1 \quad 0 \quad 0]'$.

$$\text{Mean} = b'\bar{x} = \bar{x}_1 - \bar{x}_2 = 0.766 - 0.508 = 0.258$$

$$\text{Variance} = b'Sb = S_{11} - S_{12} - S_{21} + S_{22} = 0.856 - 0.635 - 0.635 + 0.568 = 0.154$$

The covariance with the total variable from part (a) is $b'Sc$:

$$\text{Cov}(b'x, c'x) = b'Sc = \text{Row 1 Sum} - \text{Row 2 Sum} = 1.760 - 1.398 = 0.362$$

Problem 3.20

Given the dataset of 25 pairs ($n = 25$).

(a) Calculate via summary statistics

First, sum all x_1 and x_2 values to find means:

$$\sum x_1 = 292.9 \implies \bar{x}_1 = \frac{292.9}{25} = 11.716$$

$$\sum x_2 = 424.4 \implies \bar{x}_2 = \frac{424.4}{25} = 16.976$$

Calculate sums of squares and cross-products:

$$\sum x_1^2 = 4315.03, \quad \sum x_2^2 = 9021.70, \quad \sum x_1 x_2 = 4861.90$$

Determine the sample covariances using $s_{ik} = \frac{\sum x_i x_k - n \bar{x}_i \bar{x}_k}{n-1}$:

$$\begin{aligned} s_{11} &= \frac{4315.03 - 25(11.716)^2}{24} = \frac{4315.03 - 3431.6164}{24} = \frac{883.4136}{24} = 36.8089 \\ s_{22} &= \frac{9021.70 - 25(16.976)^2}{24} = \frac{9021.70 - 7204.6144}{24} = \frac{1817.0856}{24} = 75.7119 \\ s_{12} &= \frac{4861.90 - 25(11.716)(16.976)}{24} = \frac{4861.90 - 4972.2704}{24} = \frac{-110.3704}{24} = -4.5988 \end{aligned}$$

For the difference $d = x_2 - x_1$, defined by vector $a = [-1 \quad 1]'$:

$$\text{Mean} = a' \bar{x} = \bar{x}_2 - \bar{x}_1 = 16.976 - 11.716 = 5.260$$

Calculate the variance $a' S a$:

$$\text{Variance} = s_{11} - 2s_{12} + s_{22} = \frac{883.4136 - 2(-110.3704) + 1817.0856}{24} = \frac{2921.24}{24} = 121.7183$$

(b) Calculate from individual differences

Sum the individual values $d_j = x_{j2} - x_{j1}$ for all $j = 1..25$:

$$\sum d_j = (13.7 - 12.5) + (16.5 - 14.5) + \cdots + (25.6 - 13.0) = 131.5$$

$$\text{Mean} = \bar{d} = \frac{131.5}{25} = 5.260$$

Calculate the sum of squares of these differences:

$$\sum d_j^2 = (1.2)^2 + (2.0)^2 + \cdots + (12.6)^2 = 3612.93$$

Calculate the variance:

$$\text{Variance} = \frac{\sum d_j^2 - n(\bar{d})^2}{n-1} = \frac{3612.93 - 25(5.260)^2}{24} = \frac{3612.93 - 691.69}{24} = \frac{2921.24}{24} = 121.7183$$

Comparison: The results match the summary statistics found in part (a).